

**DIPLOMA EXAMINATION IN
ENGINEERING/TECHNOLOGY/MANAGEMENT/COMMERCIAL
PRACTICE**

HOSPITAL SYSTEMS
(Maximum Marks : 100)

TIME : 3HRS

PART – A

(Maximum mark:10)

Answer all Questions in one or two sentences. Each question carries **2 marks**.

I.

- 1) Define threshold of perception?
- 2) Define voltage regulation?
- 3) List the three types of gases used in hospital?
- 4) Define Doppler effect?
- 5) State any one application of bio telemetry system?

5x2=10 marks

PART – B

(maximum mark:30)

Answer any **FIVE** of the following questions. Each question carries **6 marks**.

II.

- 1) Describe electromagnet interference filter?
- 2) List the characteristics of theater lighting system?
- 3) Explain the operation of glucometer?
- 4) Describe the basics of video conferencing?
- 5) Explain the physiological parameter adaptable to biotelemetry?
- 6) Summarize enzyme electrode?
- 7) State the importance of incinerator?

5x6=30 marks

PART – C

(maximum mark:60)

Answer one full question from each Unit. Each full question carries 15 marks.

UNIT I

- III** a) Explain the working of power supply system? 8
b) Summarize the physiological effect of electric current in human body? 7

OR

- IV.** a) Illustrate the wiring details in hospital with diagram? 15

UNIT II

- V.** a) Elaborate the operation of centralized vacuum system in a hospital? 8
b) Explain single chamber autoclave with diagram? 7

OR

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|---|---|
| VI a) Explain about liquid oxygen plant in hospital? | 8 |
| b) State the importance of incinerator? | 7 |

UNIT III

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| VII a) Explain the principle and operation of speech audiometry ? | 8 |
| b) Elaborate the construction of enzyme electrode? | 7 |

OR

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| VIII a) Describe the principle and operation of pure tone audiometry? | 8 |
| b) Elaborate the working of membrane electrode? | 7 |

UNIT IV

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| IX a) Explain the physiological parameters adaptable to biotelemetry? | 8 |
| b) State the applications of telemedicine? | 7 |

OR

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| X a) Describe single channel telemetry system with block diagram? | 8 |
| b) Explain the basics of video conferencing? | 7 |

15x4=60